

3. Row 2 Column 1 = Row 3 Column 1 = 0

Row 3 Column 2 = 0

∴ The condition that for each leading entry, the column elements below it are all 0 is True.

4. From 2, we have seen that all the leading entries are 1

5. Row 1 Column 2 = 0. ∴ The condition that each leading 1 is the only nonzero entry in its column is True.

∴ The matrix is in reduced echelon form

pivot in ref: $\begin{bmatrix} \textcircled{1} & 0 & -1 & 2 \\ 0 & \textcircled{1} & 2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ pivot in original matrix: $\begin{bmatrix} \textcircled{1} & 2 & 3 & 4 \\ 4 & \textcircled{5} & 6 & 7 \\ 6 & 7 & 8 & 9 \end{bmatrix}$

pivot columns: 1 and 2

4. $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 3 & 5 & 7 & 9 \\ 5 & 7 & 9 & 1 \end{bmatrix}$ Step 1: Row 2 = Row 2 - 3 × Row 1 $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & -4 & -8 & -12 \\ 5 & 7 & 9 & 1 \end{bmatrix}$

Step 2: Row 3 = Row 3 - 5 × Row 1 $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & -4 & -8 & -12 \\ 0 & -8 & -16 & -34 \end{bmatrix}$ Step 3: Row 2 = Row 2 × $(-\frac{1}{4})$ $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 1 & 2 & 3 \\ 0 & -8 & -16 & -34 \end{bmatrix}$

Step 4: Row 3 = Row 3 + 8 × Row 2 $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & -10 \end{bmatrix}$ Step 5: Row 3 = Row 3 × $(-\frac{1}{10})$ $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

6: Row 1 = Row 1 - 7 × Row 3 $\begin{bmatrix} 1 & 3 & 5 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$